

AES GALOIS/COUNTER MODE (GCM): ENCRYPT-THEN-MAC

Encryption (CTR Mode)

Authentication (GHASH)

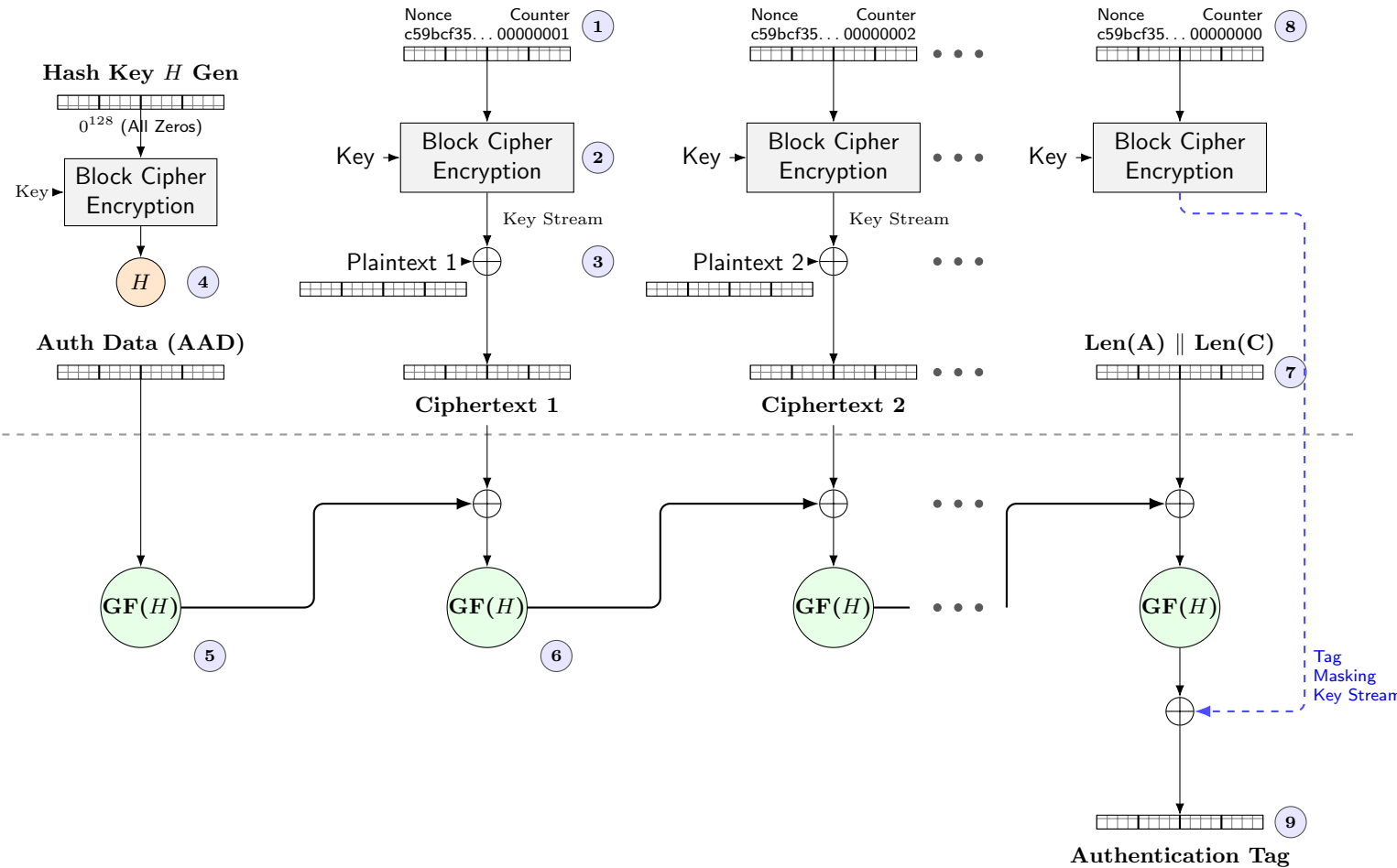


Diagram detailing AES-GCM (Galois/Counter Mode) authenticated encryption process.

PROCESS EXPLANATIONS

Hash Key Generation

- ④ **Hash Key (H):** Derived by encrypting a 128-bit block of all zeros. Used for all Galois Field multiplications.

Encryption (CTR Mode)

- ① **Data Counter:** The counter sequence for data blocks strictly begins at 1.
- ② **Key Stream Gen:** Encrypting the counter block.
- ③ **Ciphertext:** Plaintext XORed with Key Stream.

Authentication (GHASH)

- ⑤ **AAD Processing:** Auth Data is directly multiplied by H in the Galois Field $GF(2^{128})$.
- ⑥ **GHASH Chain:** Each Ciphertext block is XORed with the previously accumulated state, then multiplied by H .
- ⑦ **Lengths Block:** 64-bit representations of the AAD and Ciphertext bit-lengths cap the chain.

Tag Generation

- ⑧ **Tag Masking:** Counter 0 is encrypted to generate a unique masking key stream for the tag.
- ⑨ **Final Tag:** The final GHASH output is XORed with the mask to produce the Authentication Tag.